NexusOBE

A Neuro-Symbolic Framework for Automated Assessment Generation

Dr.A.Chandrasekar,

Professor ,

Department of CSE ,

Nandha College of Technology

1. Mahamutha Fathima.N (732123104063),

2. Mohana Prasath.K (732123104069),

3. Srihari.P.V (732123104105)

Abstract

- Preparing university question papers is currently a very slow and stressful manual task for teachers.
- Teachers waste hours trying to manually balance strict educational rules and format document tables.
- Public AI tools like ChatGPT are unsafe for confidential exams and struggle to follow exact college blueprints.
- NexusOBE solves this by combining AI with strict mathematical rules to generate perfect, ready-to-print exams automatically.
- The system saves massive amounts of time, ensures complete data privacy, and makes college accreditation audits effortless.

Introduction

- Drafting university question papers requires following strict educational standards like NBA and ABET.
- Manually balancing these requirements takes hours and often leads to human errors.
- Standard AI tools cannot solve this because they struggle with complex math, break document formatting, and risk leaking confidential data.
- NexusOBE introduces a custom software platform built specifically for colleges to overcome these exact hurdles.
- The system aims to replace manual typing with an intelligent, secure workspace that automatically generates compliant exams.

Existing System

- Current educational software acts as a basic digital filing cabinet that randomly selects pre-typed questions from an existing database.
- These systems require massive manual data entry and struggle to create brand-new, complex engineering questions on their own.
- Semi-automated tools rely on simple keyword extraction but fail to align questions with strict Course Outcomes (COs).
- Using generic AI tools (like ChatGPT) introduces severe data privacy risks and requires heavy manual formatting in MS Word.
- Existing setups cannot dynamically adapt to real-time teacher feedback or format complex exam patterns.

Existing System Advantages

- Uses basic random sampling formulas to pick questions and balance difficulty levels.
- Able to extract keywords from uploaded textbooks or syllabi to map basic topics.
- Capable of formatting the final selected questions into a standard PDF document.
- Provides secure, role-based logins for basic user access control.
- Allows for manual "Human-in-the-Loop" digital checklists so teachers maintain direct control over verification.

Existing System Disadvantages

- Highly manual mapping processes lead to cognitive exhaustion and frequent human errors when balancing Course Outcomes and Bloom's Taxonomy.
- Older systems require massive, hand-typed databases and struggle to generate brand-new, complex engineering questions.
- Relying on public cloud AI (like ChatGPT) introduces severe data privacy risks and potential leaks of confidential exams.
- Generic AI tools lack a strict mathematical rule engine, failing to guarantee exact compliance with college exam blueprints.
- Current automated systems often have a narrow scope, generating only basic memory recall questions rather than advanced analytical case studies.

Proposed System

- The proposed system utilizes a Neuro-Symbolic Rule Engine that combines artificial intelligence with strict mathematical constraints.
- A local AI engine runs securely on the institutional network to guarantee zero leakage of confidential exam data.
- Graph and Vector databases automatically map syllabus topics, balance Bloom's Taxonomy levels, and prevent question repetition.
- Retrieval-Augmented Generation (RAG) is used to read college textbooks and instantly synthesize accurate grading rubrics.
- A custom split-screen studio automatically injects and formats the generated questions directly into an official college template.

Proposed System Advantages

- Reduces exam drafting time from nearly two hours to under five minutes.
- Guarantees zero data leakage by processing confidential exam materials locally.
- Eliminates human error by mathematically enforcing strict educational blueprints.
- Removes the need for manual copy-pasting by automatically formatting the document.
- Instantly synthesizes accurate grading rubrics and creates secure, locked PDF files.

Literature Survey

- **Paper Title:** *Automated Question Generation for Electronics Engineering Exams Using Retrieval-Augmented Generation*
- **Publisher and Year:**IEEE Conference Publication (April 2026)

Key Notes from the Paper:

- Advanced Large Language Models and vector databases can successfully generate new engineering exam questions by reading raw textbooks.
- The major weakness of this framework is its absolute reliance on public cloud AI servers, creating severe data privacy risks for unreleased exams.
- The model lacks an automatic rule-engine to force the AI to respect specific college blueprints or mark breakdowns.

Literature Survey

- **Paper Title:** *PLO-Driven Engineering Curriculum Framework: A HITL Workflow for CLO Mapping*
- **Publisher and Year:** IEEE Xplore (April 2026)

Key Notes from the Paper:

- This study addresses the administrative challenges of keeping university exams compliant with international accreditation audits.
- It proposes a "Human-in-the-Loop" (HITL) manual workflow where professors use digital checklists to map questions to specific Course Learning Outcomes (CLOs).
- The major flaw is that this alignment process remains highly manual and subject to human error, leading to cognitive exhaustion and accidental formatting mismatches.

Literature Survey

- **Paper Title:** *Ontological Framework for the Analysis of Outcome-Based Curriculum in Higher Education*
- **Publisher and Year:** IEEE Xplore (2025)

Key Notes from the Paper:

- The paper presents a theoretical framework that maps university curricula using complex data structures to show hidden dependencies and prerequisites.
- It successfully proves that higher education courses and overarching learning goals are deeply interconnected.
- The major weakness is that the research is purely academic; it contains no actual code, software prototype, or functional tool for teachers to use.